

Linear Phased array for Wireless Power Transfer

6.2300 Class Project: Detailed Plan Report

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Abstract—In this paper, we demonstrate our attempt to build linear phased-array antenna systems on printed circuit boards (PCBs). This project aims to prove that low-cost PCB-milling techniques may be used to build effective phased arrays which may be applied to wireless power-transfer. The phased array consists of four patch antennas, with power distributed from a NanoVNA. We demonstrate directional selectivity by performing measurements of field intensity at various angles from the transmitter.

I. INTRODUCTION & MOTIVATIONS

The aim of this project is to direct an electromagnetic wave in a particular direction. We sought to use an array of antennas to confine the beam to a narrow angle, and a fixed phase offset between each antenna to steer the beam in a desired direction. The ability to steer a narrow beam of electromagnetic radiation has applications for communications, wireless power transfer, and sensing.



Fig. 1. Starlink Internet Satellites [1]

The Starlink satellite network operated by SpaceX (Fig 1) uses phased array radio systems to transfer internet packets to and from different geographic locales [1]. In a similar vein, future space-based solar power systems (Fig 2) will require phased arrays to steer high-power beams of microwave radiation to a receiving station on the ground. [2] Steering these beams with high precision is necessary to avoid radiating populated areas.

An Analog Devices application note describes how to create a PCB-based phased array system, but for a 2D-radar application rather than a 1D linear phased array. [3] A research team in Xi'an University in China demonstrated a PCB-based



Fig. 2. Space-Based Solar Power [2]

60 GHz phased-array system that incorporated patch antennas as well as rectangular float rings to improve directional selectivity. [4] The transmission lines, splitters, and antennas were milled into a copper surface of copper-clad dielectric boards. Details of our implementation and antenna design can be found in Section II. We collected experimental data by measuring the intensity of the EM wave at different angles from the transmitter, demonstrating the directional selectivity of our antenna array. Results from our experiments can be found in Section IV.

II. APPROACH

Changing the phase of the input to certain antennas in an array makes their lobes interfere differently with the other lobes, meaning that the radiation pattern of a phased array can be shifted or "steered" by precisely changing phases. This type of device is referred to as a dynamic phased array, and it has many applications in precise sensing.

Mathematically, the radiation pattern of an antenna is represented by a gain equation. The fabricated device utilizes a patch (or microstrip) antenna, a very basic form of antenna whose gain in the electric plane is given by:

$$G_E(\theta) = G_0 \cos^2 \left(\frac{kL}{2} \sin \theta \right)$$

Where:

$$G_0 \approx \frac{4\pi A}{\lambda^2} \eta$$

k is the wavenumber, given by $\frac{2\pi}{\lambda}$
 L is the length of our antenna [5]

Adding many of these antennas in parallel on a board will create a sharp main lobe of radiation in front of the antennas (at $\theta = 0^\circ$). The generated radiation pattern of an antenna array, $G(\theta)$, is equal to our element gain $G_E(\theta)$ (equation given above) added with our array factor $G_A(\theta)$, which describes

how changes to input phase and amplitude affect the resulting gain. The array factor is typically normalized, and for a linear patch antenna array is given by:

$$AF(\theta, \Delta\Phi) = \frac{\sin(N(\frac{\pi D}{\lambda} \sin(\theta) - \frac{\Delta\Phi}{2}))}{N \sin(\frac{\pi D}{\lambda} \sin(\theta) - \frac{\Delta\Phi}{2})}$$

Where:

$$\Delta\Phi = \pi \sin(\theta)$$

if D, the distance between antennas, is $\lambda/2$
N is the number of antennas

By changing $\Delta\Phi$ according to the above equation, we can change our array factor, which changes the angle of our main lobe [6].

We completed the design of each board in Altium. The arrays were cut from PCB blanks using desktop CNC machines. The array consisted of 4 patch antennas placed in parallel, $\frac{\lambda}{2} \approx 6.25$ cm apart, with length-tuned traces connecting each one to a tree of splitters and phase shifters leading back to a shared SMA input. To generate the signal at which the antennas will operate, we used one of the NanoVNAs provided, and to measure the output using a receiver antenna connected to Port 2 of the NanoVNA. We based our antenna design on that of M. Karthick [7], which is highly optimized for 2.4GHz operation with very low bandwidth:

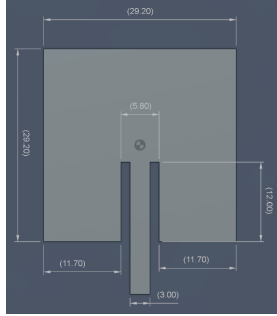


Fig. 3. Antenna Design with Dimensions (in mm) for 2.4GHz operation [7]

We utilize a Wilkinson Power Divider design that splits voltage from one-input between two voltage outputs, where we specified its dimensions to work at 2.4 GHz. The system ensures that each antenna is shifted 90 degrees from its adjacent antennas. The system consists of two identical boards with 90° phase-shifted antennas, which are themselves fed by a splitter board that produces a 180° phase shift between its outputs.

The following block diagram shows the signal transmission and reception system for this project. :

III. LEGALITY, SAFETY AND ETHICS

A. Legality

1) *Antenna Structure*: Our phased antenna array is designed to produce and transmit radio waves, which is considered an intentional radiator under Code of Federal Regulations (CFR). Title 47 Telecommunication §15.203 indicates that intentional

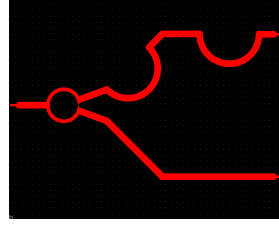


Fig. 4. Wilkinson Divider to two paths with a half-wavelength phase difference (180-degree phase shift)

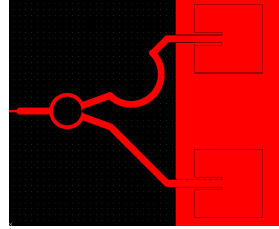


Fig. 5. Two-antenna PCB design with quarter-wavelength phase difference (90-degree phase shift)

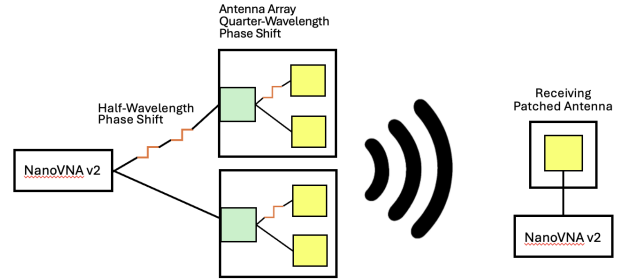


Fig. 6. Block Diagram for Phased Array

radiators with antennas must be designed to prevent users from substituting higher-gain antennas. Our antenna array is implemented through PCB-based patch antennas that are permanently attached, which cannot be replaced by users. [8]

2) *Frequency Spectrum*: FCC Part 15 Unlicensed Radio Frequency Devices restricts frequency of non-licensed transmitters from bandwidths reserved for radio communication. [9] Our target operating frequency for the antenna array is 2.4GHz. We aim to incorporate filtering and impedance matching to reduce spillage of the main transmission lobe into the restricted bands.

3) *Radiated Power*: Our design must comply with the general radiated emission limits in CFR §15.209, which restrict the electric field strength to 500 $\mu\text{V/m}$ at 3 meters, corresponding to very low radiated power. [8]

B. Safety

1) *Physical Harm to Us (Designers and Operators)*: 47 CFR § 1.1310 limits Specific absorption rate (SAR) for the general population to 0.08 W/kg averaged across the whole

body. We want to avoid over-exposure from calibrating or testing at close distance. The design complies with the maximum transmission power allowed in §15.209. [8]

2) Physical Harm to Others (Bystanders, Audience, etc.):

- *RF exposure:* During testing and demonstrations, the steerable beam may be directed toward nearby individuals. We can adopt Category One mitigation strategy under 47 CFR § 1.1307, INFORMATION: notify bystanders while the beam is active and never direct it towards people. [8]
- *Steering Errors:* Operation errors could cause the beam to point in unintended directions and expose individuals. We can avoid this by carefully testing different steering angles before demonstration.

C. Ethics

While our 2.4 GHz design operates at low radiated power, our stakeholder analysis discusses non-physical harms of systems with higher frequency and power such as 5G and RF sensing.

1) Direct Stakeholders:

- *End users (intentional)* may experience spatial variation in performance due to beamforming, which can be used to prioritize certain directions or users.
- *Sensed or targeted individuals (non-consensual)* within the sensing range may all be directly affected. Prior work such as BeamSpy shows that beamforming systems can infer location with accuracies on the order of tens of centimeters. [10]

2) Indirect Stakeholders:

- *Other electronic systems* may be affected by electromagnetic interference (EMI) from spurious emissions or beam misdirection. Under FCC Part 15, designers are responsible for meeting emission limits and avoiding harmful interference. [9]
- *Shared-spectrum users* may experience coexistence challenges.

IV. RESULTS

To demonstrate the basic functionality of our Wilkinson splitter, we created a dedicated splitter board with three SMA terminals. The S21 plot corresponds to the transmission from the input port to an output port, which is -3.7 dB at 2.5 GHz, and the S11 plot corresponds to reflection at the input, which dips to -30.1 dB at 2.5 GHz.

The performance of the phased antenna array is demonstrated by measuring antenna gain at different angles from the array, with peak gain occurring at 30°, as intended.

One potential limitation becomes apparent when we measure the impedance of a single patch antenna. At the frequency of maximum transmission, the patch antenna presents an impedance of 80Ω rather than 50Ω, which may result in some power being reflected back to the transmitter.

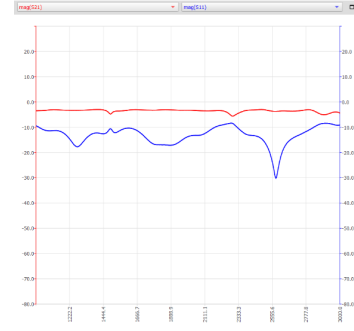


Fig. 7. Transmission and Reflection through a Wilkinson Divider

TABLE I
ANTENNA GAIN VS ANGLE AT 2.4 GHZ

Angle	Gain
0°	-46.9 dB
10°	-47.3 dB
20°	-46.2 dB
30°	-43.8 dB
40°	-44.9 dB
50°	-52.2 dB
60°	-55.4 dB
70°	-60.0 dB
80°	-62.1 dB

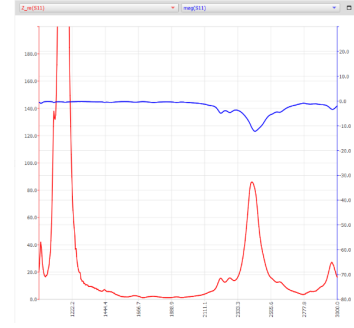


Fig. 8. High impedance of single antenna coinciding with dip in reflection coefficient

V. DISCUSSION

The antenna gain which peaks at 30 degrees is the strongest indication that our phased array is working as intended. The added 90° phase shifts between antennas result in destructive interference in the 0° axis and constructive interference along the 30° axis.

While we did simulate our Wilkinson Splitters in HFSS, we did not simulate the phase shifters, or perform experiments to validate that they are producing the correct 90° phase shift. In a future iteration of the project, it would be beneficial to simulate the both the phase shifters and antennas to validate both their impedances and the combined radiation pattern.

VI. CONCLUSION

Our project demonstrates a four-antenna patch antenna array with successful beam confinement along the 30° axis using 2-layer FR4 boards manufactured using desktop CNC machines.

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